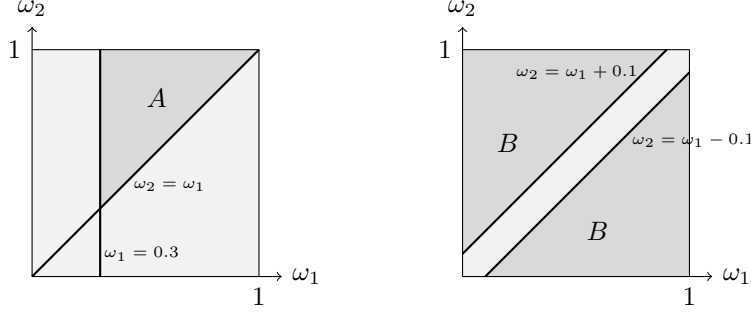


### A.2.2 Solutions

1. We illustrate the events in Figure A.2:



**Figure A.2:** The events  $A$  and  $B$  in Problem 1.

For  $A$ , note that  $A$  is the intersection of  $\{\omega : \omega_1 \geq 0.3\}$  and  $\{\omega : \omega_2 \geq \omega_1\}$  (and  $\Omega$ ). For  $B$ , note that  $|\omega_1 - \omega_2| \geq 0.1$  is equivalent to

$$\omega_2 \leq \omega_1 - 0.1 \quad \text{or} \quad \omega_2 \geq \omega_1 + 0.1.$$

The ability to visualize sets such as the above ones is important when dealing with continuous random vectors.

2. They are  $\emptyset, \{R, G\}, \{B\}, \Omega$ . Note that these events do *not* exhaust all subsets of  $\Omega$ . To the eyes of the observer the sets  $\{R\}$  and  $\{G\}$  are not admissible events.
3. Let  $X$  denote the total number of  $+$  (it is a random variable). We look for all events  $A$  which can be specified in terms of  $X$ . Since  $X$  takes values in  $\{0, 1, 2\}$ , the following events exhaust all possibilities:
  - $\emptyset$  (it nonoccurrence is known even without observing  $X$ ).
  - $\{--\} = \{X = 0\}$ .
  - $\{+-, -+\} = \{X = 1\}$ .
  - $\{++\} = \{X = 2\}$ .
  - $\{--, +-, -+\} = \{X \in \{0, 1\}\}$ .
  - $\{++, +-, -+\} = \{X \in \{1, 2\}\}$ .
  - $\{--, ++\} = \{X \in \{0, 2\}\}$ .
  - $\Omega = \{X \in \{0, 1, 2\}\}$  (always occurs).

There are 8 events. Note that  $8 = 2^3$  where 3 is the number of possible values of  $X$ . All other events, such as  $\{--, +-\}$  (the event that the second toss gives  $-$ ), require information not contained in  $X$ .

Another note: for any  $B \subset \mathbb{R}$  (even when  $B$  contains points not in  $\{0, 1, 2\}$ ) the event  $\{X \in B\}$  must be equal to one of the above.

4. We argue by contradiction. Suppose a “uniform distribution”  $\mathbb{P}$  on  $\mathbb{N}$  exists. Then the probability  $\mathbb{P}\{i\}$  does not depend on  $i \in \mathbb{N}$ . We denote the common value by  $c$ . Note that  $c \in [0, 1]$ .

Since  $\mathbb{N} = \bigcup_{i=1}^{\infty} \{i\}$ , by the sum rule we have

$$1 = \mathbb{P}(\mathbb{N}) = \sum_{i=1}^{\infty} \mathbb{P}\{i\} = \sum_{i=1}^{\infty} c.$$

Now the last summation must be either zero (when  $c = 0$ ) or infinity (when  $c > 0$ ). This contradicts the assumption that  $\mathbb{P}(\mathbb{N}) = 1$ . So there is no uniform distribution on  $\mathbb{N}$  (and hence on any countably infinite sample space).

5. Write  $A = (A \cap B) \cup (A \setminus B)$  and  $B = (A \cap B) \cup (B \setminus A)$  which are disjoint unions. By the sum rule, we have

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \setminus B) \quad \text{and} \quad \mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A).$$

By definition, the symmetric difference is given by

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

By monotonicity, we have

$$\mathbb{P}(A \setminus B) \leq \mathbb{P}(A \triangle B) \quad \text{and} \quad \mathbb{P}(B \setminus A) \leq \mathbb{P}(A \triangle B).$$

Since  $\mathbb{P}(A \triangle B) = 0$  by assumption, we have  $\mathbb{P}(A \setminus B) = \mathbb{P}(B \setminus A) = 0$ . Hence

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) = \mathbb{P}(B)$$

as desired.

6. Suppose  $A \perp\!\!\!\perp B$  and consider  $\mathbb{P}(A^c \cap B) = \mathbb{P}(B \setminus A)$ . Since  $B = (A \cap B) \cup (B \setminus A)$  is a disjoint union, we have

$$\begin{aligned} \mathbb{P}(A^c \cap B) &= \mathbb{P}(B) - \mathbb{P}(A \cap B) \quad (\text{sum rule}) \\ &= \mathbb{P}(B) - \mathbb{P}(A)\mathbb{P}(B) \quad (A \perp\!\!\!\perp B) \\ &= (1 - \mathbb{P}(A))\mathbb{P}(B) \\ &= \mathbb{P}(A^c)\mathbb{P}(B). \end{aligned}$$

So  $A^c \perp\!\!\!\perp B$ . The converse ( $A^c \perp\!\!\!\perp B \Rightarrow A \perp\!\!\!\perp B$ ) can be shown by the same argument, or more simply, by using the proven implication with  $A$  replaced by  $A^c$ .

7. The computation can be performed in R as follows:

```
# continued
p <- 0.4
p_omega <- (p^X)*((1 - p)^(n - X)) # P(omega) for all omega

pmf <- numeric(1 + n) # from 0 to n
names(pmf) <- 0:n
for (i in 0:n) {
  pmf[1 + i] <- sum(p_omega[X == i])
}

pmf # view pmf of X
dbinom(0:n, n, p) # compare with built-in function
```